DEVOPS

Principles, CI/CD lifecycle, and tool comparison

Research presentation | Beginner-friendly overview

What is DevOps?

A culture and engineering approach—not just a collection of tools

Brings development, operations, security, and product teams into shared responsibility.

Uses automation, collaboration, measurement, and fast feedback to improve delivery.

Aims for both speed and reliability; frequent releases should not mean careless releases.

Tools enable the practices, but culture and process determine whether they work.

How we got here

Waterfall → Agile → DevOps (broad evolution, not a strict replacement)

Waterfall: sequential phases; can struggle when feedback arrives late or requirements change.

Agile: smaller increments and frequent stakeholder feedback; often organized in short iterations.

DevOps: extends collaboration through build, release, deployment, operations, and improvement.

Organizations often combine methods; choose controls based on risk and context.

The Three Ways

Principles described in The DevOps Handbook

1. FLOW — optimize the full path from work to customer; reduce queues and hand-offs.
2. FEEDBACK — use tests, telemetry, alerts, and customer signals to find problems quickly.
3. CONTINUAL LEARNING — experiment safely, share knowledge, and improve after incidents.

Together: flow creates feedback; feedback informs learning; learning improves flow.

CI, Continuous Delivery, Continuous Deployment

Continuous Integration (CI): frequent code integration with automated builds and tests.

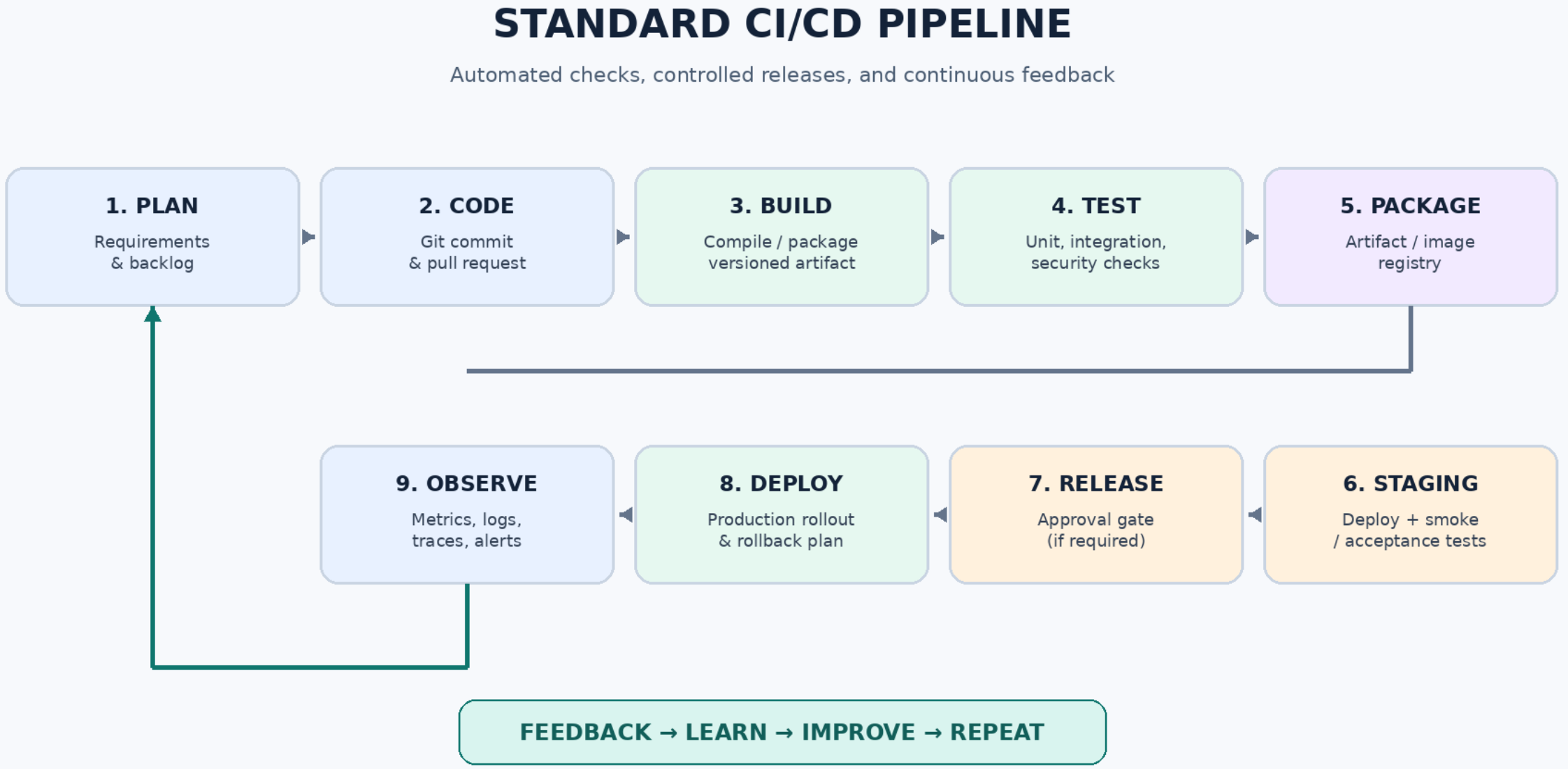
Continuous Delivery: software stays in a releasable state; production release may need approval.

Continuous Deployment: every change that passes required checks is automatically released.

Choose the level of automation appropriate to service risk, regulation, and team maturity.

A standard CI/CD lifecycle

Each stage should have a clear purpose and observable result



Tools across the lifecycle

Source and review: Git, GitHub/GitLab

CI orchestration: GitHub Actions or Jenkins

Containers and orchestration: Docker, Kubernetes

Infrastructure and configuration: Terraform, Ansible

Monitoring and visualization: Prometheus, Grafana; OpenTelemetry for instrumentation

Security: dependency, source, container, and secret scanning integrated throughout

Tool comparison: how to choose

Docker packages applications; Kubernetes orchestrates containers across clusters.

Terraform provisions infrastructure; Ansible commonly configures hosts and automates tasks.

Prometheus collects metrics; Grafana visualizes operational data; OpenTelemetry standardizes instrumentation.

No tool is best for every team: evaluate integration, security, complexity, support, and total cost.

Benefits and risks

Benefits: shorter feedback loops, repeatable deployments, earlier defect detection, better service visibility.

Risks: brittle tests, insecure credentials, over-complex toolchains, alert fatigue, and automating bad processes.

Good practice: least privilege, protected secrets, meaningful tests, staged rollout, tested rollback.

Measure both delivery and reliability: lead time, deployment frequency, change failure rate, recovery time, availability.

Conclusion

DevOps connects people, process, and technology around delivering and operating reliable software.

Agile helps teams work in increments; CI/CD automates validation and release; observability creates feedback.

The Three Ways—Flow, Feedback, and Continual Learning—guide sustainable improvement.

Start small, measure outcomes, secure the pipeline, and continuously learn.

References

Official documentation and foundational reading

Atlassian — Agile and DevOps: atlassian.com/agile/devops

Microsoft Learn — Agile development: learn.microsoft.com/en-us/devops/plan/what-is-agile-development

GitHub Actions: docs.github.com/en/actions

Jenkins: jenkins.io/doc | Ansible: docs.ansible.com

Docker: docs.docker.com | Kubernetes: kubernetes.io/docs/home

Terraform: developer.hashicorp.com/terraform/docs

Prometheus: prometheus.io/docs | Grafana: grafana.com/docs | OpenTelemetry: opentelemetry.io/docs

Kim et al., The DevOps Handbook, 2nd ed. (2021): oreilly.com/library/view/the-devops-handbook/9781098182281/